

LESSON 1

Statements and Truth

Name: _____ Date: _____

What is logic?

Logic helps you decide whether a chain of reasons actually supports a conclusion. Scientists, lawyers, programmers, and mathematicians all use the same basic toolkit — we start with the simplest piece: a **statement**.

Statements (propositions)

A **statement** (also called a **proposition**) is a declarative sentence that has exactly one truth value: it is either **true** or **false**, not both, and not “sort of.”

STATEMENTS — EACH IS CLEARLY T OR F

- “Melbourne is in Australia.” → **True**
- “The sum of 2 and 3 is 6.” → **False**
- “All prime numbers greater than 2 are odd.” → **True**
- “There are exactly seven planets in our solar system.” → depends on definition, but astronomers treat this as a fixed claim that is **false** (eight planets since Pluto’s reclassification)

NOT STATEMENTS — COMMON CATEGORIES

- **Questions:** “Is homework due tomorrow?” — asks for information; no truth value.
- **Commands:** “Show your work.” — tells someone to act.
- **Exclamations:** “That goal was incredible!” — expresses feeling; not objectively T/F.
- **Pure opinions** (without a clear standard): “Strawberry ice cream is the best flavor.” — debatable, not a fixed fact in logic class.
- **Paradoxes** (skip for now): “This sentence is false.” — breaks the true/false rule.

Borderline cases: “The test was hard” sounds like an opinion, but if we agree on a rule (“hard = more than half the class scored below 60%”), it could become a statement. In logic, we need a *clear* standard before we call something true or false.

Labeling statements with letters

We use lowercase letters as nicknames:

- Let p = “It is raining.”
- Let q = “The ground is wet.”

On a dry sunny day, p is **false** and q might also be **false**. After a storm, both might be **true**. The letters stay the same; the world decides T or F.

Open sentences vs statements

An **open sentence** contains a variable and is not a statement until the variable is fixed.

OPEN VS CLOSED

Open: “ $x + 5 = 12$ ” — true when $x = 7$, false when $x = 0$.

Closed (statement): “ $7 + 5 = 12$ ” → **True**.

Open: “Student n passed the exam.” — depends on which student.

Closed: “Maria passed the exam.” → one fixed T or F.

KEY IDEA

Algebra and geometry use statements all the time. “If $x = 4$, then $x^2 = 16$ ” is a true conditional. Logic gives the grammar for linking such sentences correctly.

Atomic vs compound (preview)

A single statement like “Snow is white” is **atomic** (one piece). “Snow is white **and** grass is green” is **compound** — built from smaller parts. Lesson 2 covers the builders (AND, OR, NOT).

Practice — Lesson 1

Difficulty: **WARM-UP** → **STANDARD** → **CHALLENGE**

1.1 **WARM-UP**

For each sentence, write **Statement (T/F)** or **Not a statement**. If it is a statement, give its truth value.

1. $15 - 8 = 7$
2. Close the window.
3. Every square has four sides.
4. Who invented the telephone?
5. Wow, that was close!

YOUR ANSWERS

1.2 **WARM-UP**

Rewrite each **statement** using letters: define $p, q, \dots$ then write “ p is true” or “ q is false” for today’s situation (you may invent reasonable T/F).

1. The school bell rings at 3:00 p.m.
2. I am wearing a hat right now.

DEFINITIONS + TRUTH VALUES

1.3 STANDARD

Classify each as **open sentence** or **statement**. If open, give one value of the variable that makes it true and one that makes it false.

1. n is an even integer.

2. $2 + 2 = 4$

3. $x^2 = 9$

4. No dogs can fly.

YOUR WORK

1.4 STANDARD

Sort these into two lists: **Statements** and **Not statements**. For statements, mark T or F.

"Please sit down." · "The Pacific Ocean is the largest ocean on Earth." · "How many states does Australia have?" · " $5 \times 0 = 0$." · "This paragraph contains exactly ten words." · "I hope it doesn't rain."

TWO LISTS + T/F

1.5 STANDARD

Turn each open sentence into a **true** statement by replacing the variable with a specific value, and into a **false** statement with a different value.

1. $a + 3 > 10$
2. The city of C is the capital of France.

TRUE VERSION / FALSE VERSION FOR EACH

1.6 CHALLENGE

Write **two** of your own examples in each category: (a) true statement, (b) false statement, (c) not a statement. Swap with a partner and check each other.

YOUR SIX EXAMPLES

1.7 CHALLENGE

Consider: “The next coin flip will land heads.” Is this a statement in classical logic? Explain your reasoning in 3–5 sentences. (There is more than one reasonable classroom answer — focus on whether the truth is fixed *right now*.)

YOUR EXPLANATION

